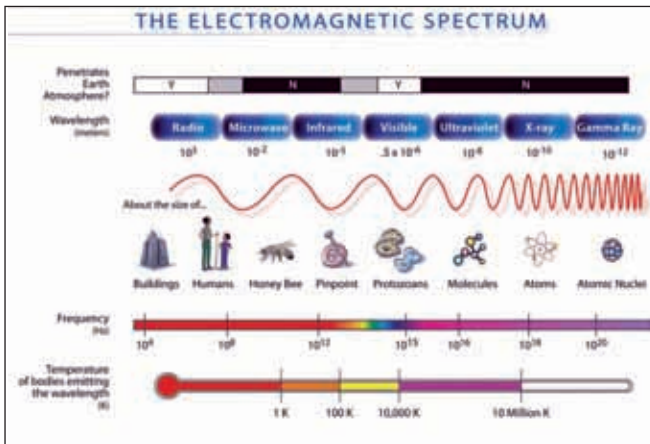




The electromagnetic spectrum. Wireless communication applications usually use frequencies lower than (wavelengths longer than) the visual spectrum.

principle is that a current carrying conductor sets up a magnetic field around it that is proportional to the current.

2.2 Frequency spectrum and its use

Electromagnetic waves are almost single-handedly responsible for all wireless technology applications today. Whether it be short range or long range, over the earth's surface or through satellites, carrying voice, images or text, all wireless information ranging from radio communication to wireless LAN are carried by electromagnetic waves. The amplitude, frequency, and the type of the signal will vary according to applications. But the fundamental concept remains the time varying electric and magnetic fields.

The frequencies used for communication purposes are usually low. This is due to the fact that higher frequencies are generally harmful for human beings, among other factors.

2.3 Modulation

Modulation is the process of shaping of a carrier wave to convey a signal. A digital message can be represented by an analogue wave by a process known as keying. There are multiple methods